

國立成功大學
測量及空間資訊學系
碩士論文

臺灣交通標誌的半自動影像標註方法
Semi-automatic Image Annotation Method
for Traffic Sign in Taiwan

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中文摘要

隨著智慧運輸系統的快速發展，自駕車系統成為產官學界共同研究與開發的目標，同時為了建置與維護自駕車用高精地圖，以影像偵測為基礎的交通標誌辨識逐漸成為熱門的研究議題。目前已有許多文獻在公開數據庫訓練交通標誌辨識模型，也在實際道路影像的驗證下，得到高準確度的辨識能力，證實基於影像的交通標誌辨識技術是可行的方案，但這些方法都必須建立在數據庫充足的條件下，目前在臺灣沒有足夠龐大具交通標誌標記的道路影像數據庫，只能利用人工標記的方式產生有限的訓練資料，即使辨識技術已十分成熟也無法被妥善使用。有鑒於此，本論文提出一種半自動化交通標誌的影像標註方法，透過對交通標誌的觀察，使用少量的標註影像來訓練 Mask R-CNN，使其依照形狀來偵測交通標誌，再經由影像優化及影像比對來更新類別，此方法只在一開始需要少量的標註影像，之後便能自動化的進行，產生具有交通標誌真實類別的標註數據，協助臺灣建立交通標誌數據庫。透過真實資料的實驗評估，本研究所提出的方法能夠有效提升交通號誌標記的效率，進而大幅改善交通號誌辨識的準確度。

關鍵詞：自駕車系統，半自動標註，物件辨識，深度學習，臺灣交通標誌數據庫。

Abstract

With the rapid development of Intelligent Transportation Systems, autonomous vehicle systems have become the goal of joint research and development by industry, government, and academia. At the same time, to build and maintain High-Definition Map for autonomous vehicles, traffic sign recognition based on image detection has gradually become a popular research topic. In recent years, many studies have been proposed to train a model to detect traffic signs based on public road image databases and obtain a high accuracy on recognition capability under the verification of actual road image, which proves that it is a practical solution to detect traffic signs using the image-based model. However, these methods must be established under enough images in databases. There is currently no large enough road image database with traffic signs in Taiwan, and only limited training data can be generated by manual annotation. Even the detection method is already prepared, it cannot be applied directly. In this thesis, we propose a semi-automatic image annotation method for traffic signs. Through the observation of traffic signs, a small number of annotated images are used to train Mask R-CNN to first detect the shape of a traffic sign. Then, the real label of a traffic sign is predicted by image enhancement and matching. The proposed method only requires a small number of annotated images initially, and the annotated data with traffic sign labels can be automatically generated to establish Taiwan traffic sign databases. Through the experimental evaluation of a real dataset, the proposed method can effectively improve the efficiency of traffic sign annotation and greatly increase the accuracy of traffic sign detection.

Keyword: Autonomous vehicle system, semi-automatic image annotation, object detection, deep learning, Taiwan traffic sign database.

Chapter 1 Introduction

1.1 Background

With the development of technology, the demand for automation is also increasing. Of course, driving is no exception. If driving can be automated, many problems caused by human driving can be eliminated, such as human's sight range is limited, and there will be distractions due to fatigue or other irresistible physiological conditions. Besides, it can also reduce many environmental costs. For example, autonomous vehicle can take passengers to their destinations and go to far-away parking lots alone, without causing a lack of parking lots in certain areas. Autonomous vehicle can bring a lot of convenience. Therefore, in recent years, many people have invested in the development of autonomous vehicle. In addition to individual enterprises, governments of various countries also actively participate and assist. However, autonomous vehicle's development is always not focused on the vehicle itself but the system. Once the autonomous vehicle system is perfect, it can be used on different vehicles to complete different services. In addition to ordinary vehicles for personal use, it can also enable mass transportation to fully automated.

Taiwan has established a High Definition Maps Research Center¹ in recent years. The primary purpose is to create high definition maps (HD maps) for automated vehicles. The information required for HD maps can be collected by vehicles with sensors while driving. The central platform will then organize and process them, and finally, upload all the data to the cloud to provide dynamic download and use of autonomous vehicles. The information required for HD maps includes traffic signs.

¹ <http://www.hdm.geomatics.ncku.edu.tw/>

1.2 Motivation

In real situations, there is still much room for improvement in the autonomous vehicle system. Among them, how to make the autonomous vehicle judge the road conditions ahead is a big challenge. If it is only to avoid pedestrians, vehicles, or unexpected objects, it can install some sensors like LiDAR to detect whether there is an object ahead. However, to further enable autonomous vehicle to "see" traffic signs is another big challenge.

In the past, many related studies have successfully used images to establish a system to detect traffic signs. The concept is to process image feature values. The most intuitive method is image feature comparison methods, such as the Scale-Invariant Feature Transform (SIFT) [17]. However, for an autonomous vehicle system, the sign's position is also required, and the image feature comparison method cannot obtain accurate position information. Therefore, in recent years, some studies have been solved through supervised learning. There are many methods, such as one of the machine learning models: Support Vector Machines [4]. You can also use the popular model in recent years: Convolutional Neural Network (CNN) [13]. Regardless of the method used, you need to have a considerable amount of data. If you want to make the system more complete, you need a more massive database. In many existing studies, usually use some public databases, such as GTSDb 22[11], GTSRB [25], and BelgiumTS dataset [18]. Most of these databases come from European, American countries, or China. Many signs are not available in Taiwan, or some of Taiwan's signs are not being contained. In the database, Taiwan does not have a related database. Therefore, if you want to establish a traffic sign recognition system suitable for Taiwan, you must first solve how to build the database.

In the above several public databases, the available information includes images and annotations. Annotation refers to the location, size, and shape information of traffic signs in the image. The annotations are all manual. The annotating images process includes collecting

images, mark the traffic signs from the image, and then giving the category name, as shown in Figure 1-1. In the whole process, the most time-consuming thing is to mark the traffic sign by clicking on the image and then finding the sign's name. Unless you know all the signs very well, it will take a lot of time. The final problem is the amount of data. If you want to use the current CNN models to achieve real-time recognition effects, such as YOLO [21] and Mask R-CNN [10], etc., it takes about one to two hundred images to train one category. If you want to achieve higher accuracy, you need more images. There are about 100 types of traffic signs in Taiwan, and some signs are not common. It is very difficult to build a complete and balanced database.

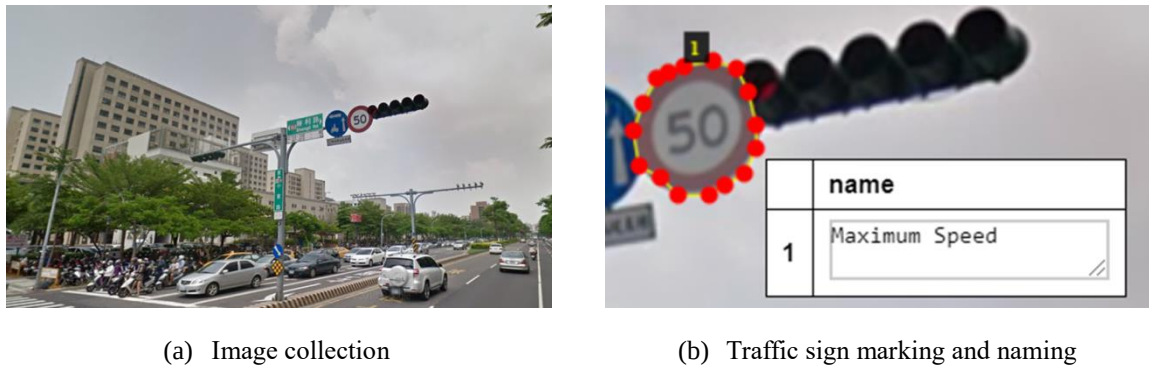


Figure 1-1 Image annotation

1.3 Research Purpose

This section describes the purpose of our research. First of all, this thesis intends to solve the problem of Taiwan's establishment of a traffic signs detecting system without a relevant database. We propose a method to improve the accuracy of CNN under insufficient data and unbalanced conditions. Furthermore, it can store the output results to assist in establishing Taiwan's traffic sign database. Our method is inspired by the observation of traffic signs. We found that the shapes of traffic signs, usually in two or three types, so we train a Mask R-CNN to detect traffic signs according to the shape. Then according to the output of the model, through specially adjusted image enhancement and image comparison

to update the category, the final output is the region of the traffic sign on the image and the real category of the sign. This method only requires a small amount of annotated images at the beginning, and then it can be automated, so we call it a semi-automatic image annotation method.

Our method can work with the platform of the High-precision Map Research and Development Center. The platform originally collected data by manually annotating the collected images, which required a lot of time and human resources. Through our proposed method, it can make the whole procedure automatically. Finally, only a small amount of human resources is needed to assist in reviewing the data, as shown in Figure 1-2.

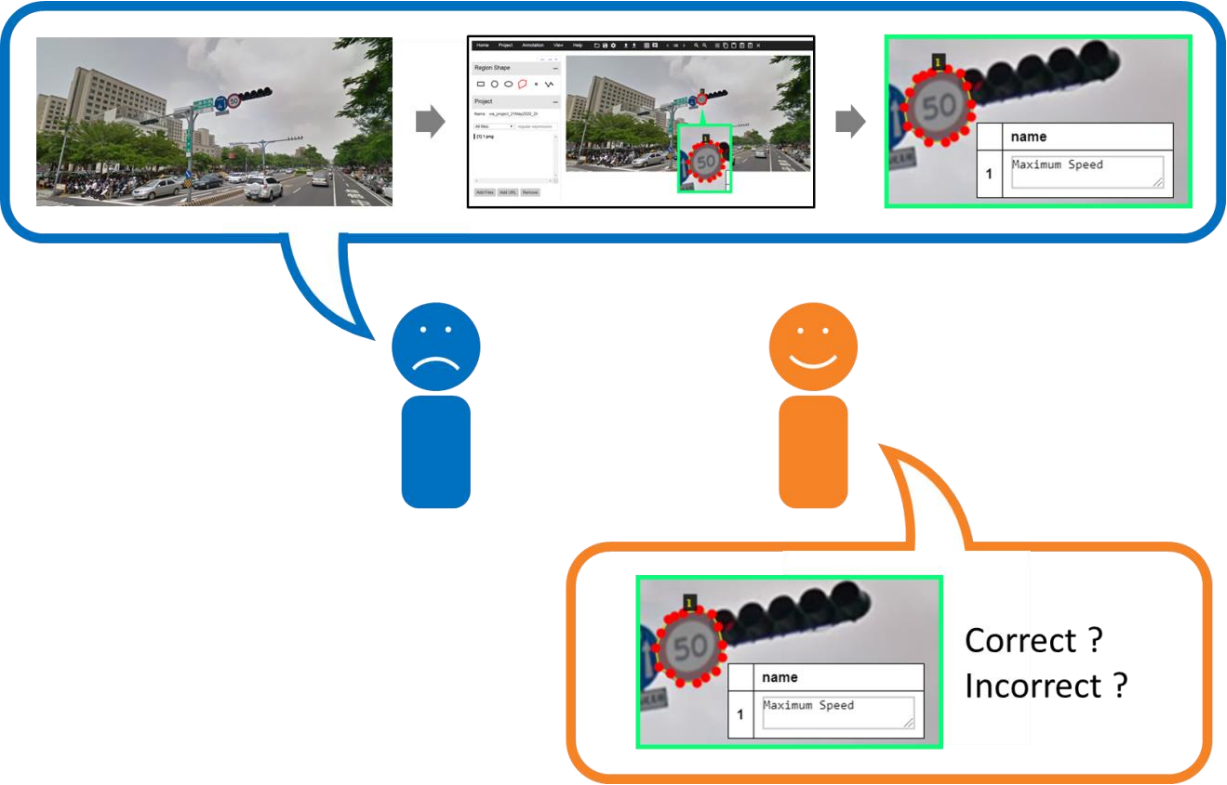


Figure 1-2 A scenario of data creating

1.4 Contribution

- (1) We detect and classify traffic signs in two stages, detect the traffic signs in the image based on their shape, and then identify their real category through image enhancement and image comparison.
- (2) In the experiment, we use the data collected in the real world to prove that the trained model is almost unusable under the insufficient and unbalanced data. Our method has significantly improved the accuracy by about 37%.
- (3) We have proposed a semi-automatic image annotation method, which can help build a Taiwan traffic signs database.
- (4) The model used in this method is Mask R-CNN, and the output is the shape of the sign, which can be converted into a bounding box for use by networks such as YOLO.

1.5 Organization

The following content of this thesis is arranged as follows. Chapter 2 reviews some methods of using images to detect traffic signs and related databases. Chapter 3 introduces our semi-automatic image annotation method. Chapter 4 is the experiment, including the testing and tuning of our method and the comparison with the intuitive method. Chapter 5 is the conclusion, and Chapter 6 is future work.

Chapter 2 Related Work

In this chapter, we review some methods of using image recognition and detection of traffic signs. Since our method is to use CNN [13], we divide this chapter into two parts. The first part is the methods before CNN is widely used, and the second part is the methods based on CNN, including YOLO [21] and Mask R-CNN [10].

Most of the methods presented in this chapter use public databases, and these databases are quite large, which bring great convenience to many studies, such as the famous German Traffic Sign Benchmark, which includes the German Traffic Sign Recognition Benchmark (GTSRB) established by Stallkamp *et al.* [25], and German Traffic Sign Detection Benchmark (GTSDB) established by Houben *et al.* [11]. GTSRB contains 43 types of traffic signs with different sizes and light conditions, a total of more than 50,000 images. GTSDB has about 900 annotated images. In [18], Mathias *et al.* established the BelgiumTS dataset, which contains 62 types of traffic signs and a total of more than 20,000 images. The Tsinghua-Tencent 100K [32] was established under consideration of more conditions. This data set contains huge illumination and weather changes, which is closer to the real situation, with 100,000 high-resolution images and 30,000 traffic signs.

2.1 Before CNN

To find traffic signs on the road, it can simply use Lidar, but Lidar cannot know the content on the signs. Therefore, the use of images to detect traffic signs has gradually become a popular research topic. Before CNN was widely used, the most intuitive approach is to use the image comparison method, such as Takaki *et al.* [27] using SIFT [17]. But to get the position of the traffic sign, the image comparison method cannot output accurate information, so many studies [12][30][8] try to use different machine learning models such as Support Vector Machines [4]. There are also many studies choices combining several different

models, such as Bahlmann *et al.* [2] combining AdaBoost and Bayesian generative models, successfully produced a system that can detect and track traffic signs.

2.2 CNN

In recent years, CNN has developed from image classification to image detection. It can not only identify objects but also detect the position of objects in the image, and even achieve instance segmentation. Therefore, more and more studies choose CNN as the basic model and try to Identify and detect traffic signs in the image.

2.2.1 YOLO

The famous CNN model: YOLO can detect objects in the image with a bounding box, so it has been widely used since the first version, such as Tran *et al.* [28] used YOLO to detect traffic signs in Vietnam. Artamonov *et al.* [1] used YOLO on a mobile GPU to make the entire system highly mobile. However, the accuracy of YOLO still needs to be improved, so YOLOv2 [22] appeared, and not only improved the accuracy but also increased the detection speed. Zhang *et al.* [31] took YOLOv2 as the basic structure and proposed a network with a high detection speed; the fastest can reach 0.017 seconds per image. With people's pursuit of detection speed, YOLOv3 [23] appeared as expected. YOLOv3 was inspired by the Feature Pyramid Network [14]. It uses a structure with similar concepts to detect smaller objects. Ćorović *et al.* [3] used YOLOv3 to detect five objects on the road, including cars, trucks, pedestrians, traffic signs and lights, and prove its effectiveness under different environmental conditions.

2.2.2 Mask R-CNN

The bounding box can only indicate the rough size and position of the object. For autonomous vehicles, highly accurate information will be more helpful. In HP maps, the information of the object must have an outline. Many studies have proposed CNN networks that can perform instance segmentation. For example, Mask R-CNN can output a mask with each object's outline. Mask R-CNN is the latest product of the R-CNN series. Previous methods can only output bounding boxes. First, the concept of R-CNN [7] is to use the selective search [29] on the image to select multiple proposed regions, and use CNN to extract features from the candidate regions Value, then classified by SVM. But the whole process costs much time, so Fast R-CNN [6] appeared. It directly extracted features of the entire image to speed up the time, which is different from extracting each proposed region's feature by R-CNN. Then, Faster R-CNN [24] proposed Region Proposal Network to replace selective search, making it faster. Finally, Mask R-CNN was proposed. It is a fully convolutional network that can not only detect objects but also perform instance segmentation, so it has been widely used. For example, Tabernik *et al.* [26] used Mask R-CNN as basic model and proposes some improvements to detect more kinds of traffic signs.

2.2.3 Other Models

In addition to the well-known models: YOLO and Mask R-CNN, many models are directly used or modified. Even new architectures are proposed, such as RetinaNet [15] published by Facebook AI Research and used by Rajendran *et al.* [20]. In [16], Liu *et al.* proposed a multi-scale region-based CNN called MR-CNN, and Ou *et al.* [19] proposed a feature aggregation multipath network called FAMN, both of which proposed new architectures for detecting smaller traffic signs.

Chapter 3 Methodology

In this chapter, we propose a semi-automatic image annotation method for Taiwan traffic signs. Figure 3-1 is a flowchart of our proposed method. The whole method is divided into two stages. The first stage is to use a small number of annotated images to train a Mask R-CNN model to detect traffic signs according to the shape. The second stage is to use the previous stage results to get the real category of traffic signs through image enhancement and image matching methods.

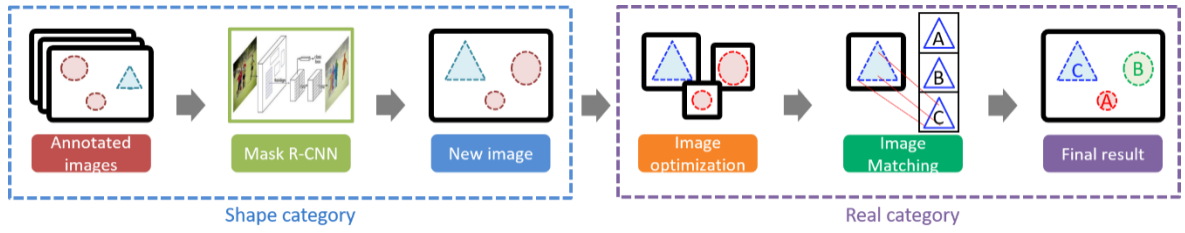


Figure 3-1 Flowchart of our method

3.1 Shape Category

The most intuitive approach to use the CNN model to detect traffic signs is to prepare training data and train the model to detect different traffic signs in the new image, as shown in Figure 3-2. However, there are about a hundred types of traffic signs in Taiwan. If CNN is used for detecting traffic signs directly, it will be a classification problem of one hundred categories. The shapes of traffic signs are mostly in two or three types, and there are no signs in irregular shapes. Therefore, we think it is relatively simple for CNN to detect traffic signs according to their shapes. At this stage, we annotate a small number of images and train a CNN model to detect traffic signs according to different shape categories.



Figure 3-2 Intuitive approach

3.1.1 Image Annotation



Figure 3-3 Image annotation

As shown in Figure 3-3, in this step, the image can be obtained by taking photos or videos. Next, use annotating tools such as VIA² to annotate traffic sign in the image, then give it a category name. The category can be given the shape of the sign and the real category name at the same time through a special naming. For example, circular signs can be named with a C and a number, and triangle signs can be named with a T and a number, such as shown in Figure 3-4. Since the first stage is to detect signs by shape, the total number of categories is reduced from three digits to single digits, and the required training materials are also significantly reduced, thus saving a lot of time.



Figure 3-4 Special naming

² <http://www.robots.ox.ac.uk/~vgg/software/via/>

3.1.2 Object Detection

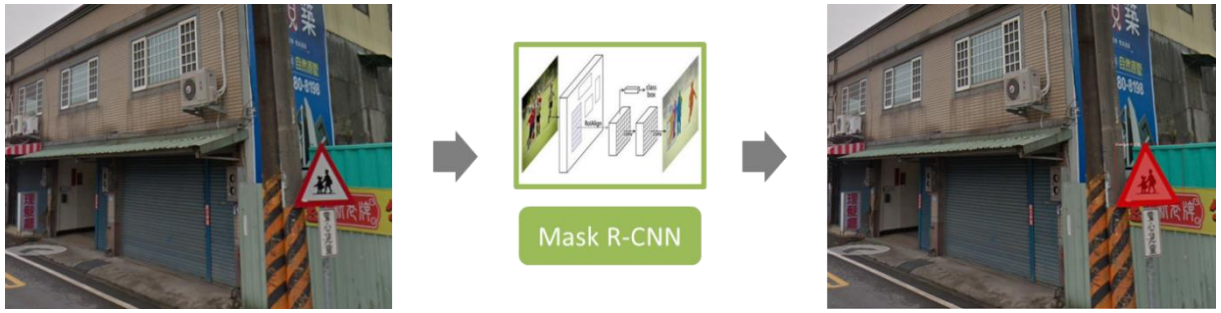


Figure 3-5 Object detection

For the High Definition Maps Research Center's information format, the object needs to have the category, position, and outline information, so we choose Mask R-CNN for model training, as shown in Figure 3-5. After the training is completed, the traffic sign can be detected according to the shape when the new image is input, as shown in Figure 3-6.



Figure 3-6 Sign detection by shape

3.2 Real Category

In the last stage, we get the mask of the traffic sign in the new image, but the category only has a shape. At this stage, we tried to update the category through image enhancement and image comparison, changing the shape category to the real category of traffic signs.

3.2.1 Image Enhancement

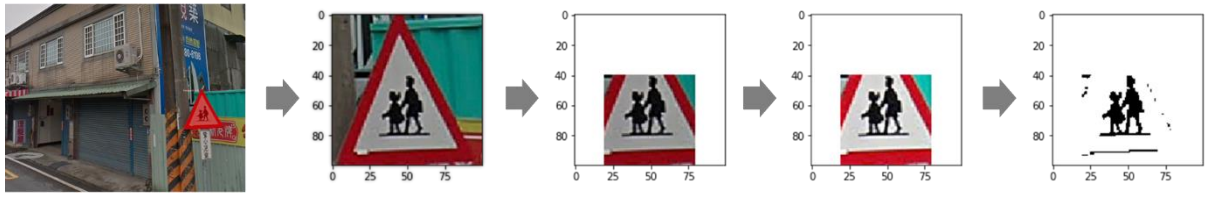


Figure 3-7 Image enhancement

According to the output result of Mask R-CNN, we extracted each traffic sign from the complete image, turned it into an individual image, and tried to get the sign's real category through image comparison. However, the image comparison method's robustness is lower than CNN, so image enhancement before image comparison can get better results. Figure 3-7 is the flow chart of image enhancement. First, because the shape of the sign is different, the important part of the sign also different. Therefore, we crop the image again in different positions according to the shapes. For example, the circular signs will crop out from the middle of the image; the triangle signs will crop out from under the middle of the image. Then we optimize the image, and there are two steps in image enhancement. At first, we tried to pick the image's color, leaving only a few specific colors. We found that the brightness of the image would affect the result through experiments, so we adjusted the brightness of the image before the color picking.

The first step of image enhancement is to adjust the brightness of the image. The intuitive approach is to use histogram equalization [9]. By expanding the commonly used brightness, the histogram distribution is more balanced, and the contrast of the image will be more obvious. However, through experiments, we found that histogram equalization is easy to make the background more visible. Therefore, the approach we use is to calculate the number of pixels that are too dark. If the threshold value is reached, we use gamma correction to the image. The brightness histogram of the image will move to the right. The image brightness will increase.

The second step of image enhancement is color picking. Since there are still have other objects than traffic signs in the image, such as background or leaves, which easily affect the result of image matching. We try to pick the color of the image, leaving only the specific color. Through our observation of traffic signs, we decided to keep only the commonly used color of the sign: black. After all the image enhancement processes, only the black pixels in the particular areas are retained in the image, and the rest are set to white. Figure 3-8 is an example.

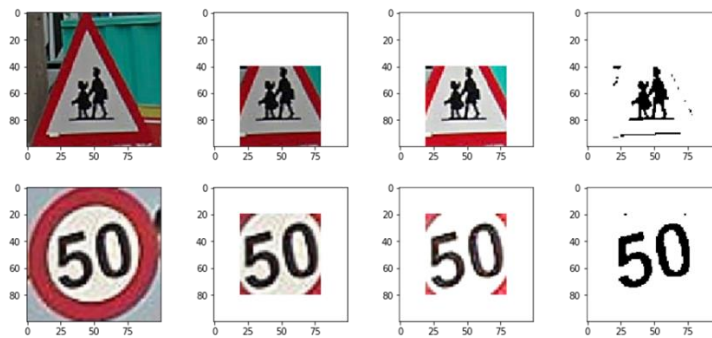


Figure 3-8 Final result of image enhancement

3.2.2 Image Matching

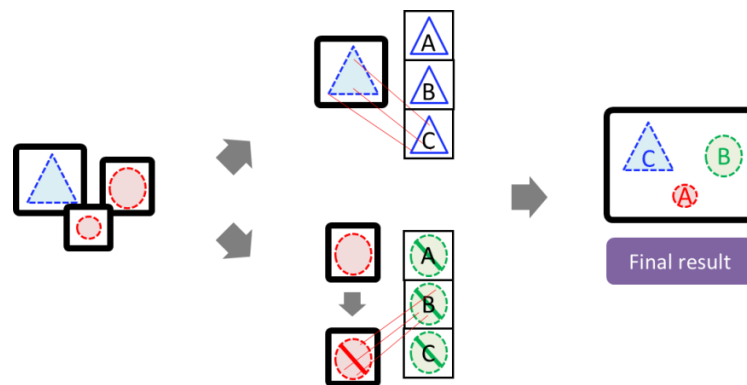


Figure 3-9 Image Matching

As shown in Figure 3-9, to get the real category without training data, we think image matching is a feasible approach, because image matching only needs an image to compare. The method we use is SIFT. SIFT is a widely used image matching method that can still work when the image is rotated or offset. The image used for comparison is the original legend of the traffic sign, as shown in Figure 3-10. Since the circular signs can be divided

into two types: with or without a slash. We can first calculate the number of red pixels in the center, as the first stage of classification. Then select the matching legend according to whether there are slashes, as shown in Figure 3-11.

Through experiments, we found that the result of image matching using a single legend is not good enough. Hence, we add the homography method to filter out some of the most unlikely matches. Finally, we tried to combine several legends into one image according to the shape category and then determine the result based on the location of the feature points of the image matching. Regardless of which image is used for image matching, in addition to finding the legend with the most feature points, several rules must be reached to determine the real category of signs. Otherwise, it will be regarded as impossible to determine. The rules for determining the real category are as follows: (1) Feature points need to be at least more than 5 points to exclude signs that are not in the recognition range, or other objects with similar shapes. (2) The difference between the largest and the second number of feature points must be greater than 3. It is to eliminate the situation when the category cannot be clearly determined, even if the feature points are enough.



Figure 3-10 Traffic sign legend

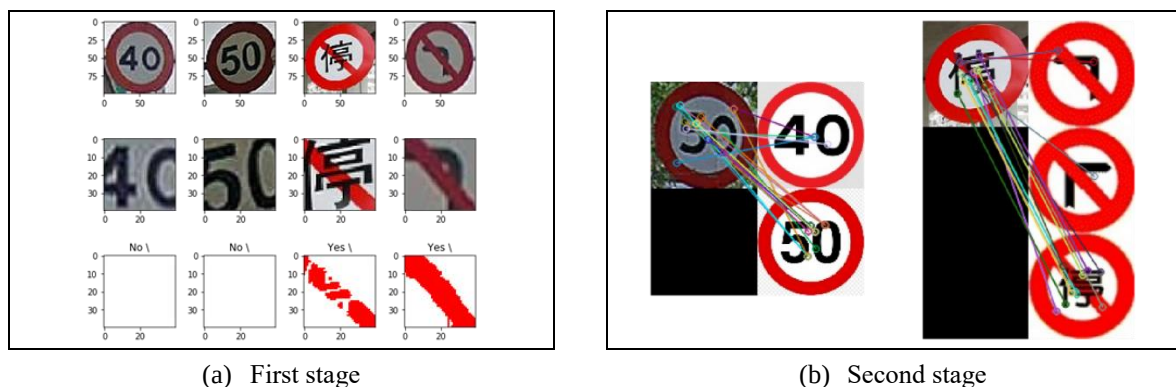


Figure 3-11 Circular signs classification

Chapter 4 Experiment

In this chapter, we first explain the data set and evaluation used in the experiment. The main experiment is divided into two parts. The first part is our method testing and adjustment to determine the best configuration of different variables. The second part is the comparison with the intuitive approach. All experiments are implemented with Python 3.6 Tensorflow-gpu-1.12, GPU is GeForce RTX 2080-Ti, and the operating system is Linux Ubuntu 18.04.

4.1 Dataset

We use Google Maps³ to collect road images with traffic signs in Taiwan-Taoyuan, Taichung, and Tainan. The training data has 367 images, and the test data has 97 images. The shape of Taiwan's traffic signs includes quadrilateral, circles, and triangle. The content of quadrilateral signs is mostly written descriptions and will have different content according to the area where it is placed, cannot be completely divided into several categories. So our data set contains only 2 types of shapes: circle and triangle, which contains several easy-to-collect categories, with 5 and 6 real categories, respectively. The number of times each category appears in the data set is shown in Table 4-1. The evaluation of the experiment is the mAP with different IOU.

Table 4-1 Dataset

category dataset											
Train	79	90	31	42	16	58	50	21	12	24	43
Test	10	10	10	10	10	10	10	10	10	10	10

4.2 Method Tuning

The first part of the experiment includes the test of the accuracy of the shape category and the impact of different variables on the real category's accuracy. The variables are divided into two parts: image enhancement and image matching.

³ <https://www.google.com.tw/maps>

4.2.1 Accuracy of shape category

This experiment is mainly to verify that Mask R-CNN can correctly detect traffic signs based on the shape categories. We divided the data set into two categories according to their shapes: circle and triangle, and train on Mask R-CNN. Finally, we test the accuracy according to different IOU settings, as shown in Table 4-2.

Table 4-2 Accuracy of shape category

IOU	mAP (%)
0.50	97.94
0.70	97.94
0.75	97.94
0.80	95.36
0.85	82.39
0.90	48.75

4.2.2 Variables for Image Enhancement

This experiment is mainly to test the effect of image enhancement variables on the results. The variables include using histogram equalization (HE) or gamma correction to adjust the image brightness, and whether to use color picking (CP). IOU is set at 0.8. All results are as shown in Table 4-3.

Table 4-3 Variables for image enhancement

HE	Gamma	CP	mAP (%)
		V	52.06
V			45.79
V		V	44.67
	V		51.75
	V	V	54.12

4.2.3 Variables for Image Matching

This experiment is to test the effect of the image matching method on the results. It is divided into using a single legend, using a single legend with the homography method, and using a combined legend. The image used gamma correction and color picking for the enhancement, and IOU is set at 0.8. From the experimental results, using the combined legend for image matching will be better than the single one. Results are as shown in Table 4-4.

Table 4-4 Variables for image matching

legend	mAP (%)
single	20.33
single+ homography	28.01
combined	54.12

4.3 Comparison of Methodology

The final experiment is our method and intuitive approach, comparing the results with different IOU settings. The intuitive approach is to use Mask R-CNN to detect and classify traffic signs directly. After experiments with different variables, our method is finally configured as follows: use gamma correction to adjust image brightness, use color picking, and use combined legends in image matching. All results are as shown in Table 4-5.

Table 4-5 Comparison of methodology

IOU	mAP (%)	
	ori	ours
0.50	19.47	56.19
0.70	19.47	56.19
0.75	19.47	56.19
0.80	19.47	54.12
0.85	18.96	47.94
0.90	13.34	34.77

Chapter 5 Conclusion

In this thesis, we propose a method to detect the traffic signs without a large number of image data. The main purpose is not to directly use on the autonomous vehicle system, but to assist in the establishment of the required database. Due to the traffic signs contain more than one hundred categories, the proposed method consists of two stages. The first stage is to detect traffic signs by shape category based on Mask R-CNN. The model training only requires less than 400 images, which can significantly reduce the time cost for training data preparation. In the second stage, through image enhancement and image matching, the category can be updated from the shape to the real category. Even if there are still some errors, the work of manually annotating images has been replaced with an easier data checking because the entire process is automatic. The overall time spent can be reduced significantly. In addition, since the model we used is Mask R-CNN, the output is the mask with the objects' outline. By finding the maximum and minimum values of x and y coordinates, it can be converted into a bounding box, which becomes the format required by models such as YOLO.

Chapter 6 Future Work

In future work, we expect to improve the current method further. First, we plan to improve the accuracy of image matching by adding more steps in image enhancement. Next, because some traffic signs contain numbers, and the numbers are different depending on where the signs are placed. We plan to recognize the numbers on traffic signs through a simple CNN or other methods. Besides, we also consider designing more complex method structures to handle more special cases. For the object detection model, we also consider more ways to increase the accuracy, including using continuous images as input, using a generative adversarial network to generate training data, or using an existing database to train the model and then fine-tune it. Finally, since our method is to assist in the establishment of a traffic sign database, we expect that the proposed method can iteratively run and add the generated data for traffic sign detection model training. After a period of learning time, the accuracy of traffic sign detection in real categories can be significantly improved. Then, there can be more applications, such as analyzing the features trained by the model to find out where traffic signs are not easily detected and providing a reference for the government when designing them.

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